

- 6 (a) A coil with 500 turns is placed in a uniform magnetic field of flux density $5.0 \times 10^{-2} \text{ T}$. The area of the coil perpendicular to the field is $2.5 \times 10^{-2} \text{ m}^2$, as shown in Fig. 6.1.

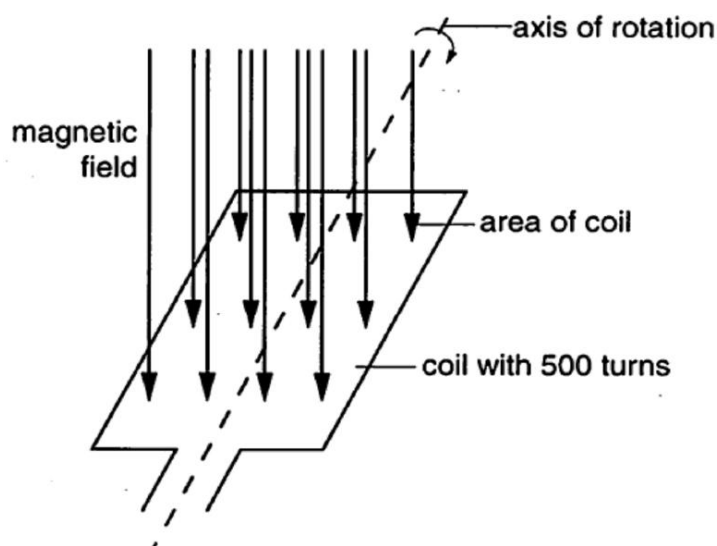


Fig. 6.1

Calculate the magnetic flux linkage of the coil. Give an appropriate unit.

magnetic flux linkage = unit: [1]

Solution:

$$\Phi = NBA \cos \theta = (500)(5.0 \times 10^{-2})(2.5 \times 10^{-2}) \cos 0^\circ$$

$$= 0.625 = 0.63 \text{ Wb}$$

where θ : angle between the normal to the plane of the coil and the magnetic field.

OR

$$\Phi = NBA \sin \alpha = (500)(5.0 \times 10^{-2})(2.5 \times 10^{-2}) \sin 90^\circ$$

$$= 0.625 = 0.63 \text{ Wb}$$

where α : angle between the plane of the coil and the magnetic field.

1 mark for both correct answer & units.

1

- (b) The coil in (a) is rotated at a constant angular velocity about the axis in Fig. 6.1. The flux linkage Φ of the coil varies with time t , as shown in Fig. 6.2.

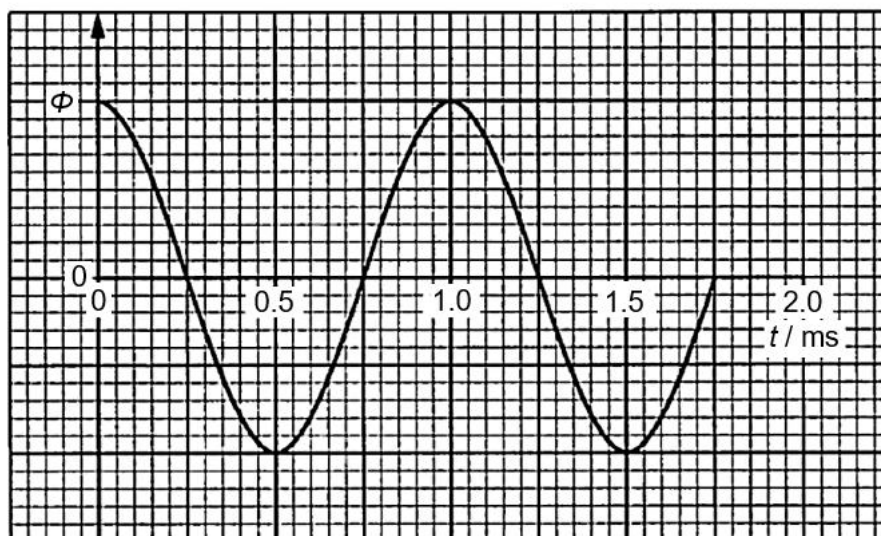


Fig. 6.2

- (i) Calculate the maximum induced electromotive force (e.m.f.).

maximum induced e.m.f. =V [2]

Solution:

From graph,

$$\Phi = \Phi_0 \cos(\omega t)$$

$$E = -d\Phi/dt = -\omega\Phi_0 \sin(\omega t) = -(2\pi/T)\Phi_0 \sin(2\pi/T)t$$

$$= -\left(\frac{2\pi}{1.0 \times 10^{-3}}\right)(0.625) \sin\left[\left(\frac{2\pi}{1.0 \times 10^{-3}}\right)t\right]$$

Hence **Maximum induced e.m.f.**,

$$E_0 = \omega\Phi_0 = \left(\frac{2\pi}{1.0 \times 10^{-3}}\right)(0.625)$$

$$= 3927.0 = 3900 \text{ V (2 s.f.)}$$

1
1

		(ii)	Calculate the root-mean-square value of the induced e.m.f.	
			root-mean-square e.m.f. =V	[1]
			Solution: Since sinusoidal AC, $E_{\text{rms}} = E_0 / \sqrt{2} = 3927.0 / \sqrt{2} = 2776.8 = 2800 \text{ V}$	1
		(iii)	Explain why the flux linkage changes <i>sinusoidally</i> as the coil is rotated.	
				
				
				
				[1]
			Solution: The effective area of the coil perpendicular to the magnetic field to vary sinusoidally with time. OR The magnetic flux density perpendicular to the plane of the coil vary sinusoidally with time.	1 (1)

7 Fig. 7.1 shows some of the energy levels of an helium atom